



ECTO Glossary

Term	Definition
Assumption sensitivity	The capacity to notice, identify, and evaluate assumptions that influence how a theory, model, argument, or empirical test works.
Auxiliary assumption	An additional assumption needed to derive, interpret, or test a theory's implications, but that is not usually one of the theory's central claims.
Boundary condition	A condition that specifies when, where, or for whom a theory's claims are expected to hold, weaken, change, or fail.
Capability	A broad, transferable capacity that supports effective, adaptive performance across diverse theoretical tasks, domains, and unfamiliar situations.
Capacity	An ability or disposition of an entity or system to do something or produce an outcome under suitable conditions.
Causal reasoning	Reasoning about whether and how changes in one entity, process, or condition produce changes in another.
Coherence	The extent to which the elements of a theory fit together consistently and without contradiction.
Competency	The integrated ability to combine relevant knowledge, skills, capabilities, judgment, and resources to perform a substantive theoretical function to an adequate standard.
Computational model	A formal model implemented as an executable algorithm or simulation that specifies how inputs, states, or processes generate outputs over time.



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Conceptual precision	The capacity to formulate and distinguish concepts, claims, and relations clearly enough to avoid important ambiguity or conflation.
Conditional requirement	A knowledge, skill, or capability requirement that is necessary only under particular circumstances, for particular theoretical tasks, or for particular kinds of theories.
Construct	A theoretical concept used to represent or refer to a property, state, process, mechanism, or other feature relevant to a theory.
Construct architecture	The organized set of constructs within a theory or domain, including how they are defined, differentiated, related, grouped, or hierarchically structured.
Core requirement	A knowledge, skill, or capability requirement that is broadly necessary for competent performance of a particular theoretical function.
Counterfactual expectation	A claim about what would be expected to occur under an alternative or hypothetical condition, including conditions that did not actually occur.
Counterfactual reasoning	Reasoning about what would happen, or would have happened, if some condition, event, or causal factor were different.
Cross-cutting requirement	A knowledge, skill, or capability requirement that supports competent performance across multiple theoretical tasks or competencies rather than belonging to only one.
Dependency	A relation in which successful performance of one theoretical activity, skill, or competency requires or is supported by another element.
Diagrammatic representation / model	A representation that uses a diagram to make theoretical entities, variables, processes, or relations explicit, such as a causal diagram or network.



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Empirical interface	The explicit connections between a theory and empirical research, including how theoretical concepts are measured or manipulated, what observations bear on the theory, and what empirical implications follow from it.
Epistemic judgment	The capacity to evaluate the strength, relevance, limitations, and implications of evidence when forming or revising theoretical claims.
Epistemic status	The evidential standing of a claim, proposed target, or theoretical element: how strongly it is supported and how confidently it should currently be treated.
Explanandum (plural: explananda)	Something that a theory is intended to explain. In ECTO, an explanandum may be a phenomenon, capacity, or problem.
Explanatory value	The extent to which a theory contributes to explaining or understanding its explananda.
Formal representation	An explicit representation of theoretical content using a rule-governed system such as logic, mathematics, diagrams, algorithms, or another precisely specified notation.
Formalization	The process of translating theoretical content into a more explicit and rule-governed representation so that assumptions, relations, and implications can be specified and examined more precisely.
Functional explanation	An explanation that identifies what a capacity, process, structure, or behavior does or what role it plays within a broader system without necessarily specifying the mechanism that produces it.
Hybrid representation / model	A representation that deliberately combines two or more representational forms, such as verbal propositions, diagrams, equations, or executable simulations.
Indicator	An observable or measurable feature that provides evidence about a theoretical construct.



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Index	A measure that combines information from multiple indicators according to a specified rule to represent a construct or quantity.
Integrative reasoning	The capacity to connect and coordinate information, concepts, explanations, or evidence from different sources, domains, or levels of analysis.
Inter-competency dependency	A dependency in which successful performance of one competency relies on knowledge, skills, outputs, or judgments associated with another competency.
Intervention prediction	A prediction about what will happen when a variable, process, or condition is deliberately changed.
Knowledge	The concepts, facts, principles, methods, and informational resources that a theorist must understand, be aware of, or know how to access in order to perform theoretical work.
Mathematical model / representation	A formal representation that uses mathematical expressions, such as equations, functions, or probability distributions, to specify relations and derive consequences.
Mechanistic explanation	An explanation that shows how a phenomenon is produced by specifying relevant components, their activities, and how they are organized.
Model	A deliberately simplified or structured representation of some target, theory, process, or system that can be used for purposes such as explanation, exploration, derivation, comparison, or prediction.
Model family	A set of related models that share important assumptions or structure while differing in particular specifications, parameterizations, or variants.



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Ontology	An explicit account of the kinds of entities, properties, categories, and relations assumed or distinguished within a domain.
Operationalization	The specification of concrete procedures or rules by which a theoretical construct is represented, manipulated, measured, or otherwise connected to observable evidence.
Parsimony	The extent to which a theory accomplishes its explanatory task without introducing more independent constructs, assumptions, or mechanisms than are needed.
Phenomenon	A relatively stable and general feature of the world that is supported by empirical evidence and can serve as a target of explanation.
Preliminary sketch	An early and incomplete explanatory account that identifies candidate components, functions, processes, or relations while leaving important details unspecified.
Problem	An empirical or conceptual issue, difficulty, or question that a theory is intended to address or help resolve.
Process-based explanation	An explanation that focuses on the sequence or dynamics through which an outcome develops, such as stages, transitions, feedback loops, trajectories, or time-dependent interactions.
Provisional pattern	An observed pattern that may be meaningful but is not yet sufficiently established to be treated as a robust phenomenon or explanandum.
Representational flexibility	The capacity to select, interpret, translate between, and combine different forms of theoretical representation when appropriate to the task.
Research programme	A coordinated set of research activities organized around theoretical questions, assumptions, models, or goals and designed to guide future empirical, formal, or methodological work.



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Scope	The range of explananda, systems, populations, contexts, conditions, or other circumstances to which a theory is intended to apply.
Skill	A specific, learnable ability to perform a defined cognitive, methodological, representational, communicative, or practical operation relevant to theoretical work.
Speculative target	A proposed target of explanation whose existence, recurrence, or relevant properties have not yet been sufficiently established.
Systems thinking	The capacity to reason about systems of interacting components, including organization, feedback, dependencies, dynamics, and relations across levels.
Taxonomy	A systematic classification of entities or concepts into categories, often but not necessarily organized hierarchically.
Testability	The extent to which a theory can be connected to observations or interventions that could provide evidence bearing on its claims.
Theory	A structured set of concepts, assumptions, and claims intended to describe, explain, or organize understanding of one or more explananda.
Theory diagnosis	The systematic identification of weaknesses, gaps, tensions, ambiguities, or other problems in an existing theory in order to determine what may need clarification, development, or repair.
Theoretical implication	A claim, expectation, explanation, prediction, or other consequence that follows from a theory together with relevant assumptions.
Uncertainty management	The capacity to recognize, represent, communicate, and reason appropriately under uncertainty rather than treating tentative claims as settled.



Term	Definition
Verbal theory	A theory expressed primarily in natural language that states relevant constructs, claims, relations, and conditions in verbal form.